Literature status: general relaxation in BD and recession functions

Source arXiv:1008.2089; answers arXiv:1903.05771 and 1911.01356

13 August 2026

Status: literature already answered.

Original question

Remark 6.9 of Rindler [1] (source PDF p. 44) asks for a relaxation theorem for integrands f that are not symmetric-quasiconvex, with f replaced by its symmetric-quasiconvex envelope SQf. It records two obstacles: whether the strong recession function $(SQf)^\infty$ exists was unknown, and the paper's framework could not establish lower semicontinuity after replacing it by the generalized recession function $(SQf)^\#$.

Exact later resolution

Kosiba–Rindler [2], Theorem 1.3 (supporting PDF pp. 3–4), proves the desired weak-star relaxation in the standard homogeneous setting. If $f\colon \mathbb{R}^{d\times d}_{\mathrm{sym}} \to [0,\infty)$ is continuous and

$$m|A| \le f(A) \le M(1+|A|),$$

then the relaxation from $LD(\Omega)$ to $BD(\Omega)$ is

$$\overline{F}[u,\Omega] = \int_\Omega (SQf)(Eu)\,dx + \int_\Omega (SQf)^\# \left(\frac{dE^s u}{d|E^s u|}\right) d|E^s u|.$$

Immediately afterward, the authors emphasize that this theorem requires no assumption on a recession function and explicitly cite Rindler's source paper as the preceding strong-recession result and discussion.

The later survey of De Philippis–Rindler [3] makes the logical status fully explicit. On supporting PDF pp. 25–26 it says that, in general, the strong recession function $(SQf)^\infty$ does not exist (citing a published counterexample), introduces $(SQf)^\#$ for precisely this reason, and states the same result as Theorem 4.5. Thus the source's existence question has a negative answer, while its relaxation objective is achieved by the proposed generalized-recession formula.

Provenance, scope, and search evidence

This is not an agent-derived implication: [2] cites the source as its reference [34], contrasts its Theorem 1.3 with that paper's result, and points back to the source's recession-function remarks. The survey cites the source as [64] and the relaxation theorem as [51]. The theorem quoted above assumes an x-independent continuous integrand with two-sided linear growth; it does not by itself settle arbitrary dependence on x or u. It does exactly resolve the homogeneous relaxation problem discussed in Remark 6.9, and the failure of $(SQf)^\infty$ is general.

References

[1] F. Rindler, *Lower semicontinuity for integral functionals in the space of functions of bounded deformation via rigidity and Young measures*, Arch. Ration. Mech. Anal. 202 (2011), 63–113; arXiv:1008.2089.

[2] K. Kosiba and F. Rindler, *On the relaxation of integral functionals depending on the symmetrized gradient*, arXiv:1903.05771 (2020 version).

[3] G. De Philippis and F. Rindler, *Fine properties of functions of bounded deformation—an approach via linear PDEs*, arXiv:1911.01356 (2019).